NE523 Midterm Exam

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Part 1 Integro-differential Form

The one-speed, steady-state transport equation with a uniformly distributed monoenergetic isotropic source and vacuum boundary on square slab with side length a is

$$\left[\mu\frac{\partial}{\partial x}+\eta\frac{\partial}{\partial y}\right]\psi(x,y,\mu,\eta)+\sigma_t\psi(x,y,\mu,\eta)=S, \quad x\in(0,a),\quad y\in(0,a),\quad \mu,\eta\in(-1,1) \tag{1a}$$

$$\psi(0,y,\mu,n)=0,\quad \mu>0 \tag{1b}$$

$$\psi(a,y,\mu,n)=0,\quad \mu<0 \tag{1c}$$

$$\psi(x,0,\mu,n)=0,\quad \eta>0 \tag{1d}$$

$$\psi(x,a,\mu,n)=0,\quad \eta<0 \tag{1e}$$

where $\sigma_t(x,y)=\sigma_t$ is constant absorption cross section, and $S(x,y)=S$ is the uniformly distributed isotropic source ($neutrons\ cm^{-1}str^{-1}$).

Part 2 Integral Form

Since neutrons travel in a straight line (between collisions) in direction (μ,η), any position in the neutron's path can be parameterized with

$$x=x_0+\mu s \tag{2a}$$

$$y=y_0+\eta s \tag{2b}$$

where s is the total distance traveled from the point (x_0,y_0). Taking the derivative of ψ along the parameter s, using the chain rule, yields

$$\frac{\partial\psi}{\partial s}=\frac{\partial\psi}{\partial x}\frac{\partial x}{\partial s}+\frac{\partial\psi}{\partial y}\frac{\partial y}{\partial s} \tag{3}$$

Using the derivative of Equation 2, this becomes

$$\frac{\partial \psi}{\partial s} = \frac{\partial \psi}{\partial x}\mu + \frac{\partial \psi}{\partial y}\eta, \tag{4}$$

which is equivalent to the bracketed term in Equation 1a. The transport equation can be converted into an ordinary differential equation by substituting Equations 2 and 4 into the integro-differential form:

$$\frac{d}{ds}\psi(x_0 + \mu s, y_0 + \eta s, \mu, \eta) + \sigma_t \psi(x_0 + \mu s, y_0 + \eta s, \mu, \eta) = S. \tag{5}$$

Part 3 Integration Along Characteristic

Integration of Equation 5 can be facilitated using an integrating factor

$$I(s) = \exp\left[\int_0^s \sigma_t ds''\right] \tag{6}$$

where

$$\frac{d}{ds}I(s) = \sigma_t I(s). \tag{7}$$

Multiplying Equation 5 by $I(s)$ yields

$$\frac{d}{ds}\left\{\psi(x_0 + \mu s, y_0 + \eta s, \mu, \eta)\right\} I(s) + \sigma_t I(s)\psi(x_0 + \mu s, y_0 + \eta s, \mu, \eta) = SI(s). \tag{8}$$

Using Equation 7, this can be rewritten as

$$\frac{d}{ds}\left\{\psi(x_0 + \mu s, y_0 + \eta s, \mu, \eta)\right\} I(s) + \frac{d}{ds}\left\{I(s)\right\} \psi(x_0 + \mu s, y_0 + \eta s, \mu, \eta) = SI(s), \tag{9}$$

which is, by inverting product rule for derivatives, equivalent to

$$\frac{d}{ds}\left\{\psi(x_0 + \mu s, y_0 + \eta s, \mu, \eta) I(s)\right\} = SI(s). \tag{10}$$

Integrating this along the characteristic yields

$$\int_0^s ds' \psi(x_0 + \mu s', y_0 + \eta s', \mu, \eta) I(s') = S\int_0^s ds' I\left(s'\right) \tag{11}$$

$$\psi(x_0 + \mu s', y_0 + \eta s', \mu, \eta) \exp\left(\int_0^{s'} \sigma_t ds''\right)\Bigg|_{s'=0}^{s} = S\int_0^s ds' \exp\left(\int_0^{s'} \sigma_t ds''\right). \tag{12}$$

Introducing optical length $\tau(x_0 + \mu s, y_0 + \eta s, x_0, y_0) = \int_0^s ds'' \sigma_t$, this can be written as

$$\psi(x_0 + \mu s', y_0 + \eta s', \mu, \eta) \exp\left[\tau(x_0 + \mu s', y_0 + \eta s', x_0, y_0)\right] \Big|_{s'=0}^{s}$$
$$= S \int_0^s ds' \exp\left[\tau(x_0 + \mu s', y_0 + \eta s', x_0, y_0)\right]. \quad (13)$$

The left-hand side is evaluated at the bounds,

$$\psi(x_0 + \mu s, y_0 + \eta s, \mu, \eta) \exp\left[\tau(x_0 + \mu s, y_0 + \eta s, x_0, y_0)\right] - \psi(x_0, y_0, \mu, \eta)$$
$$= S \int_0^s ds' \exp\left[\tau(x_0 + \mu s', y_0 + \eta s', x_0, y_0)\right], \quad (14)$$

and the terms are rearranged,

$$\psi(x_0 + \mu s, y_0 + \eta s, \mu, \eta) \exp\left[\tau(x_0 + \mu s, y_0 + \eta s, x_0, y_0)\right]$$
$$= \psi(x_0, y_0, \mu, \eta) + S \int_0^s ds' \exp\left[\tau(x_0 + \mu s', y_0 + \eta s', x_0, y_0)\right], \quad (15)$$

finally yielding

$$\psi(x_0 + \mu s, y_0 + \eta s, \mu, \eta) = \frac{\psi(x_0, y_0, \mu, \eta) + S \int_0^s ds' \exp\left[\tau(x_0 + \mu s', y_0 + \eta s', x_0, y_0)\right]}{\exp\left[\tau(x_0 + \mu s, y_0 + \eta s, x_0, y_0)\right]}. \quad (16)$$

3(a) Solution

Since σ_t is constant, the optical length is

$$\tau(s) = \int_0^s ds'' \sigma_t = \sigma_t s. \quad (17)$$

Since the boundary condition is vacuum, then $\psi(x_0, y_0, \mu, \eta) = 0$ for appropriate choice of (x_0, y_0) corresponding with the direction of the characteristic.

For $\eta > \mu > 0$, the solution is

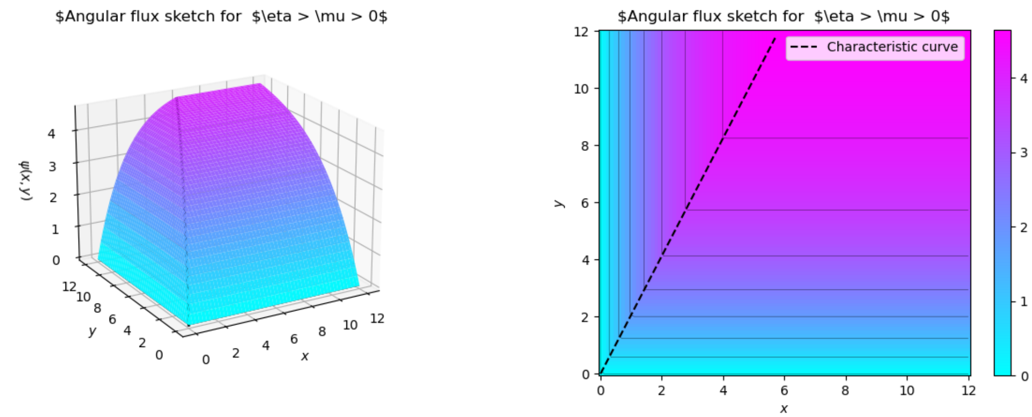
$$\psi(s, \mu, \eta) = S \frac{\int_0^s ds' e^{\sigma_t s'}}{e^{\sigma_t s}} \quad (18)$$
$$= S \frac{\frac{1}{\sigma_t}\left[e^{\sigma_t s} - 1\right]}{e^{\sigma_t s}} \quad (19)$$
$$\psi(s, \mu, \eta) = \frac{S}{\sigma_t}\left[1 - e^{-\sigma_t s}\right] \quad (20)$$

where $s = \min\{\frac{x}{\mu}, \frac{y}{\eta}\}$ is the distance from the intersection of the characteristic curve and the boundary. Then the solution for $\eta > \mu > 0$ is

$$\psi(x, y, \mu, \eta) = \begin{cases} \frac{S}{\sigma_t}\left(1 - \exp\left[-\frac{\sigma_t}{\mu}x\right]\right), & \eta x \leq \mu y \\ \frac{S}{\sigma_t}\left(1 - \exp\left[-\frac{\sigma_t}{\eta}y\right]\right), & \eta x > \mu y \end{cases} \tag{21}$$

Part 4 Solution Sketch

An example of the solution shape for $\eta > \mu > 0$ can be seen in Figure 1, and the solution along the characteristic line can be seen in Figure 2. The solution along the characteristic line asymptotically approaches the infinite-medium solution, S/σ_t (see Equation 20). This is also the solution in 1d slab geometry for $\mu = 1$.



(a) (b)

Figure 1: Example solution for $\eta > \mu > 0$

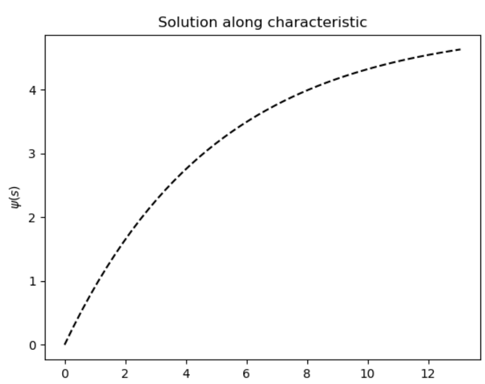


Figure 2: Solution along characteristic line

Part 5 Other Cases

For other characteristic curves, the parameterized solution (Equation 20) does not change, only the definition of s.

For $\mu < 0, \eta < 0$, (x_0, y_0) now lies on the top or right boundaries ($x = a$ or $y = a$), so $s = \min\left\{\frac{x-a}{\mu}, \frac{y-a}{\eta}\right\}$, and the solution becomes

$$\psi(x,y,\mu,\eta) = \begin{cases} \frac{S}{\sigma_t}\left(1 - \exp\left[-\frac{\sigma_t}{\mu}(x-a)\right]\right), & \eta(x-a) \le \mu(y-a) \\ \frac{S}{\sigma_t}\left(1 - \exp\left[-\frac{\sigma_t}{\eta}(y-a)\right]\right), & \eta(x-a) > \mu(y-a). \end{cases} \tag{22}$$

Similarly, for $\mu < 0, \eta > 0$, (x_0, y_0) lies on the lower and right boundaries, $x = a$ and $y = 0$, so $s = \min\left\{\frac{x-a}{\mu}, \frac{y}{\eta}\right\}$. The solution is

$$\psi(x,y,\mu,\eta) = \begin{cases} \frac{S}{\sigma_t}\left(1 - \exp\left[-\frac{\sigma_t}{\mu}(x-a)\right]\right), & \eta(x-a) \le \mu y \\ \frac{S}{\sigma_t}\left(1 - \exp\left[-\frac{\sigma_t}{\eta}(y)\right]\right), & \eta(x-a) > \mu y. \end{cases} \tag{23}$$